

Urban Growth Modeling: Tampa, Florida

CPLN6750-401

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Introduction

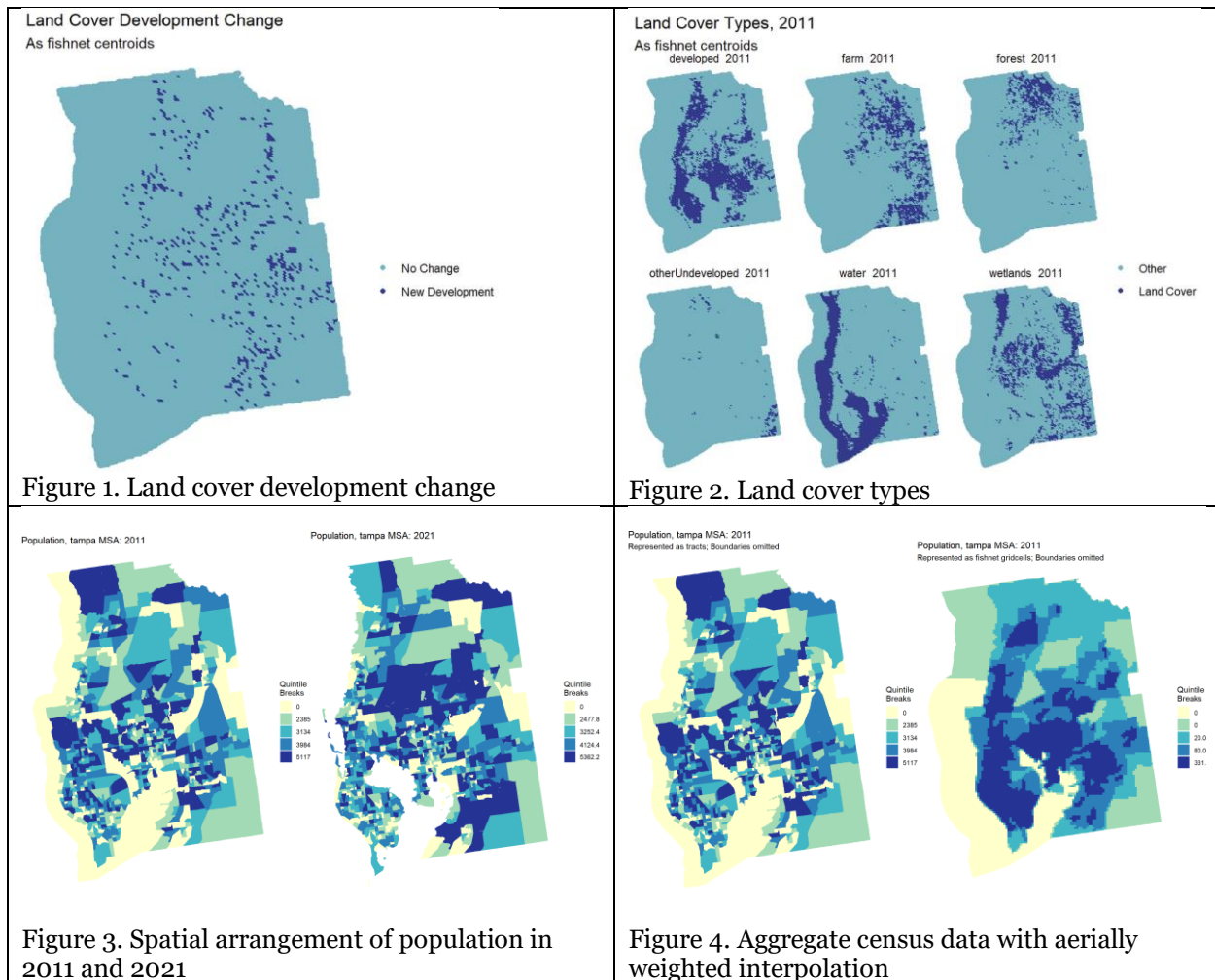
In preparation of growing development, urban sprawl, and regional transportation infrastructure needs, this forecast model predicts land use change in the Tampa, FL metropolitan statistical area (MSA) through 2031. A logistical regression model was created to predict development chances based on the trends of 2011-2021 by incorporating land cover, census population data, and existing highway systems. A new bridge was placed into the 2021 dataset, connecting downtown St. Petersburg and Ruskin, to simulate this vital infrastructure's future impact on the area's development.

Planning Rationale

The Tampa MSA has recently undergone rapid urban and suburban expansion, largely driven by a growing population, rising real estate investments and major transportation infrastructure. The growth forecast supports the region's development plan by creating a model where growth is likely to occur through 2031 based on land cover usage, population and current highways. The proposed infrastructure project is a new bridge connecting HW 375, which ends in downtown St. Petersburg to HW 674, that begins in Ruskin. This intervention was chosen to reduce roadway congestion and shorten the current 32-mile commute, which takes residents approximately 36 minutes. This additional transportation infrastructure is vital to support the population and economic development of the Tampa MSA.

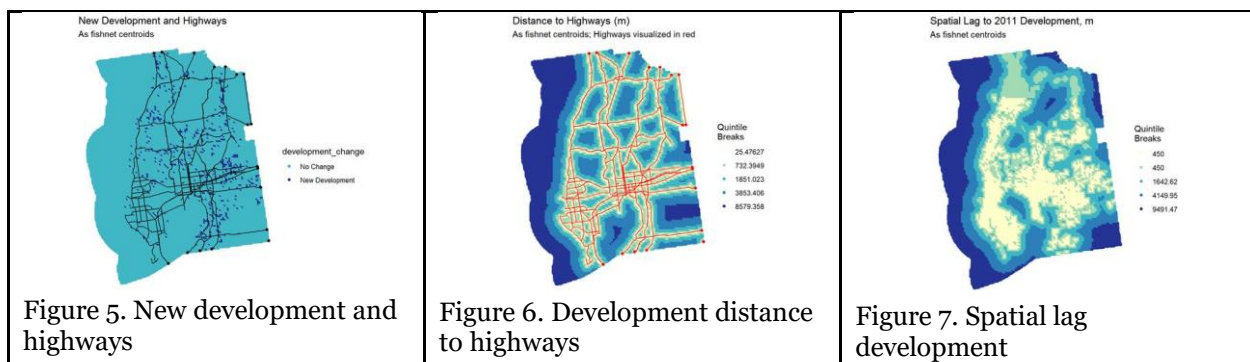
Exploratory Maps

The exploratory maps of the current land cover and relevant predictors in the Tampa MSA reveal patterns of development and population distribution across the study area. As shown in the top panels of the figure below, there is a mix of developed and undeveloped areas in the region, with the most development clustered together in the northern and central portions of the Tampa MSA. The middle and bottom panels display population distribution, showing an increase in population density in both urban and suburban zones.

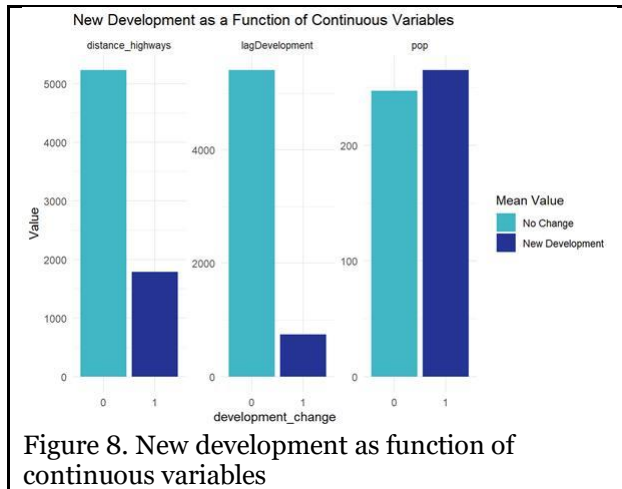


Feature Engineering

The variables chosen as predictors include distance to highway, the spatial lag of development, and census variables including population density. The hypothesis follows that the closer a cell is to highway infrastructure, the more likely it is to develop due to accessibility being a key determinant of development potential.



A bridge connecting HW 375 from downtown St. Petersburg to HW 674 near Ruskin was drawn in mapedit to represent the proposed bridge. Spatial lag of development was calculated, measuring from all fishnet centroids to the nearest 2 fishnet centroids that were developed in 2011. County boundaries were then spatially joined into our data set as a fixed effect on the models. On average, a cell that converted from undeveloped to developed was in areas closer to highways and new developments with a higher population compared to a cell that didn't convert. From 2011 to 2021, farmland was the most impacted with 1.8% converted to developed.



Land_Cover_Type	Conversion_Rate
farm	1.8%
forest	0.57%
otherUndeveloped	0.13%
wetlands	0.8%

Figure 9. Conversion rate by land cover types

Model Results

Using a Binary Logistic Regression, the data was split into 50% training and test sets. Six separate glm models were created, each of which added a new variable. Model1 includes only previous land cover types, model2 adds spatial lag, model3 adds population, model4 adds a fixed effect for county, model5 adds the distance to highway, and model6 interacts distance to highway and development lag.

```
##
## Call:
## glm(formula = development_change ~ wetlands + forest + farm +
##   otherUndeveloped, family = binomial(link = "logit"), data = datTrain)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -2.58242    0.87752  -33.314 < 2e-16 ***
## wetlands1     -1.12767    0.28639  -5.464 4.66e-08 ***
## forest1       -1.35159    0.34536  -3.914 9.18e-05 ***
## farm1         -0.18172    0.16264  -1.137  0.264
## otherUndeveloped1 -0.08481    0.46893  -0.181  0.856
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 2134.1 on 5141 degrees of freedom
## Residual deviance: 2899.8 on 5137 degrees of freedom
## AIC: 2899.8
##
## Number of Fisher Scoring iterations: 6
```

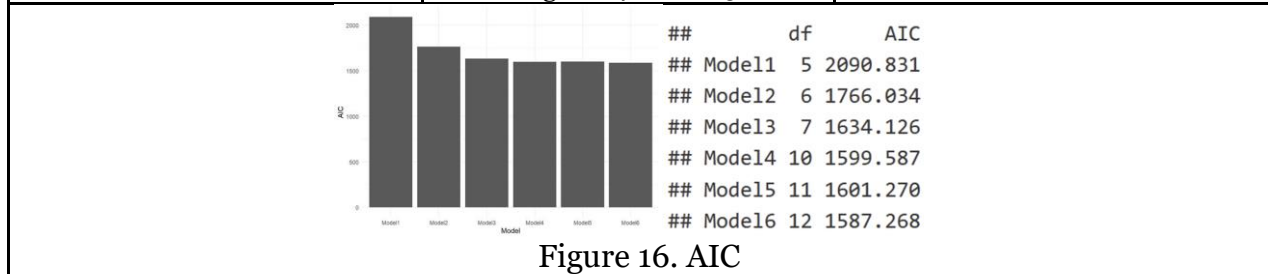
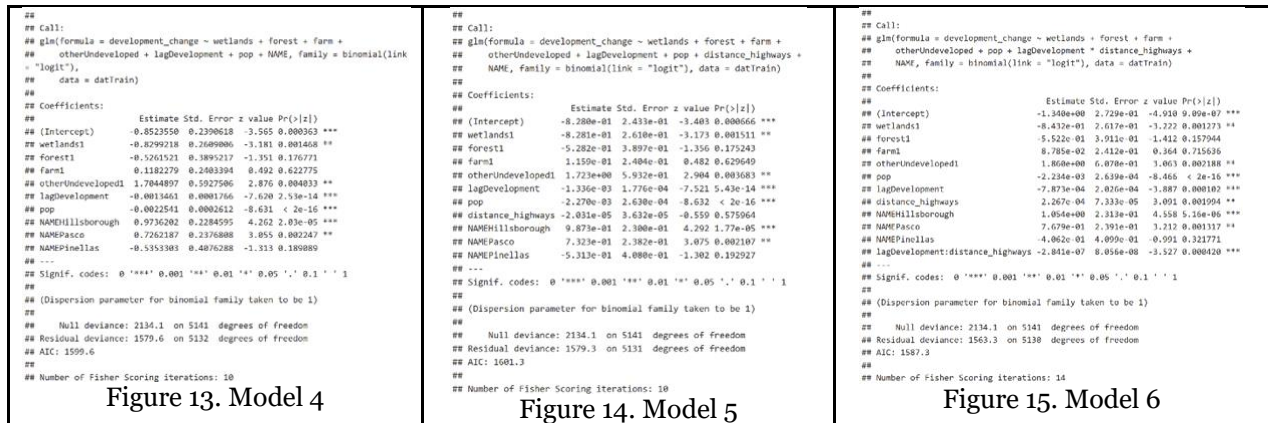
Figure 10. Model 1

```
##
## Call:
## glm(formula = development_change ~ wetlands + forest + farm +
##   otherUndeveloped + lagDevelopment, family = binomial(link = "logit"),
##   data = datTrain)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -1.455188    0.1189312  -13.133 < 2e-16 ***
## wetlands1     -0.2329164    0.2589151  -1.008  0.313
## forest1       -0.1838039    0.2794755  -0.272  0.786
## farm1         0.9368954    0.2328428    4.020 5.81e-05 ***
## otherUndeveloped1 2.2638546    0.5777332    3.919 8.91e-05 ***
## lagDevelopment -0.0013453    0.0002667   -6.866 6.59e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 2134.1 on 5141 degrees of freedom
## Residual deviance: 1754.0 on 5136 degrees of freedom
## AIC: 1766
##
## Number of Fisher Scoring iterations: 18
```

Figure 11. Model 2

```
##
## Call:
## glm(formula = development_change ~ wetlands + forest + farm +
##   otherUndeveloped + lagDevelopment + pop, family = binomial(link = "logit"),
##   data = datTrain)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   -0.1797135    0.1534998  -1.173  0.24169
## wetlands1     -0.6840748    0.2599889  -2.632 0.00849 **
## forest1       -0.7829106    0.3829669  -2.042 0.04117 *
## farm1         0.3886345    0.2387924    1.253 0.19682
## otherUndeveloped1 1.5714244    0.5848528    2.687 0.00721 **
## lagDevelopment -0.0013868    0.0001761   -7.787 6.89e-15 ***
## pop           -0.0021847    0.0002473   -8.438 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 2134.1 on 5141 degrees of freedom
## Residual deviance: 1620.1 on 5135 degrees of freedom
## AIC: 1634.1
##
## Number of Fisher Scoring iterations: 18
```

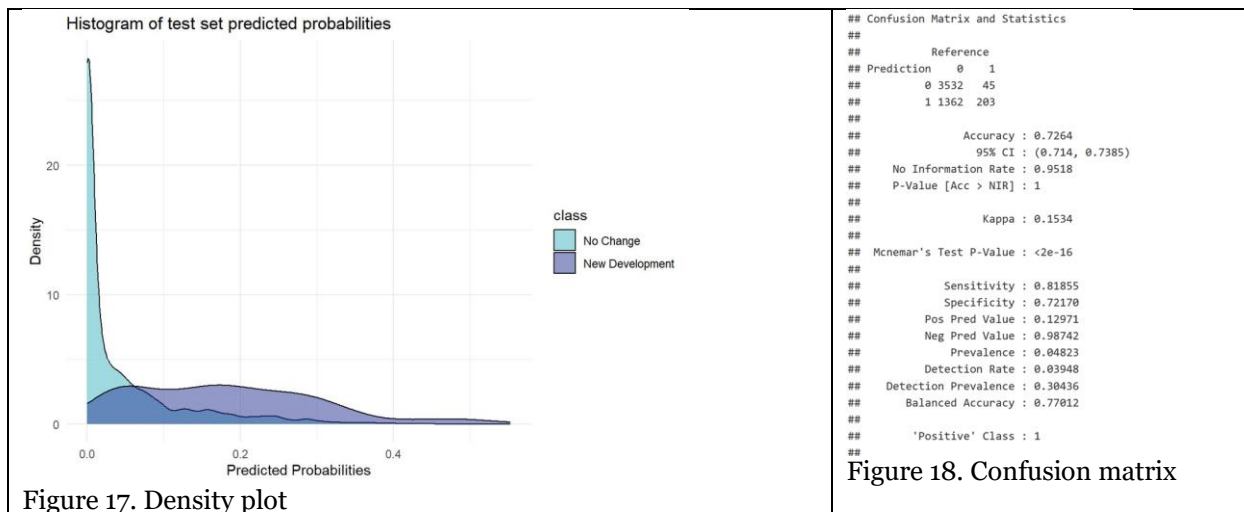
Figure 12. Model 3



Model 6 has the most variables that are significant along with the lowest AIC which indicates a better goodness of fit. Given this, the model was chosen for our analysis.

Model Validation

To validate our model, a density plot was used to find the threshold for classifying predicted probabilities will be set. The cutoff value of 0.05 was set as it is around the intersection of no development and development in the density plot below. This indicates that values above 0.05 are predicted as a 1 (new development), and below it is predicted as a 0 (no development).



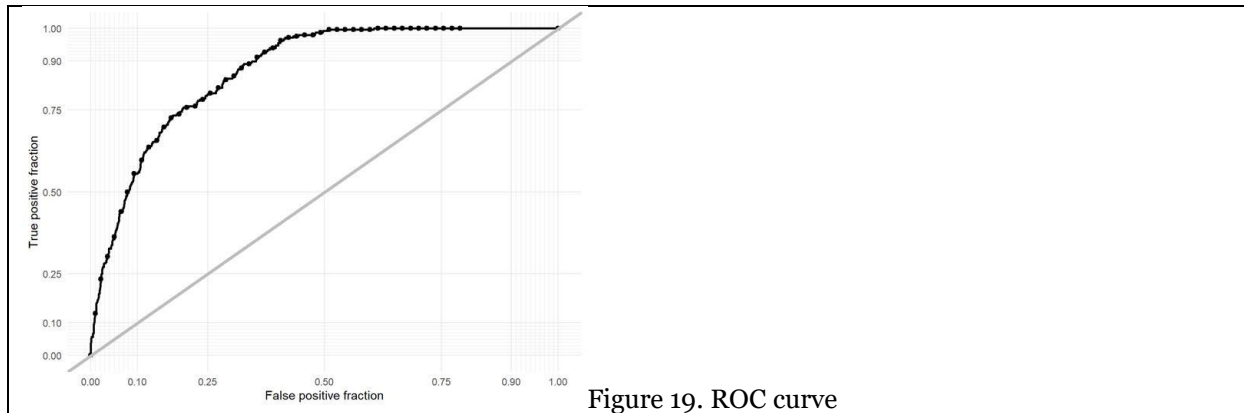


Figure 19. ROC curve

Looking at the confusion matrix, we can see that the model is highly imbalanced with very few true positives relative to the number of negatives. The ROC curve shows a “healthy” tradeoff between correctly identifying positive cases and incorrectly identifying negative cases.

Analyzing Errors

When setting the model thresholds, it was optimized for predicting for specificity. This is due to higher amounts of false positives being costly with more transportation infrastructure than necessary being implemented. The two thresholds set were 5%, reflecting the intersection in the density plot, and 17%, reflecting the highest point of new development in the density plot. The 17% threshold results have higher false negatives but are selected due to lower false positives.

NAME	Model_Type	TN_Rate_Specificity	TP_Rate_Sensitivity
Hernando	confResult_05	77.96776	88.70968
Hernando	confResult_17	97.36199	19.35484
Hillsborough	confResult_05	60.04902	88.25758
Hillsborough	confResult_17	83.85417	56.06061
Pasco	confResult_05	69.06764	90.62500
Pasco	confResult_17	86.94698	70.00000
Pinellas	confResult_05	88.02326	70.58824
Pinellas	confResult_17	97.38372	11.76471

Figure 20. Comparing thresholds

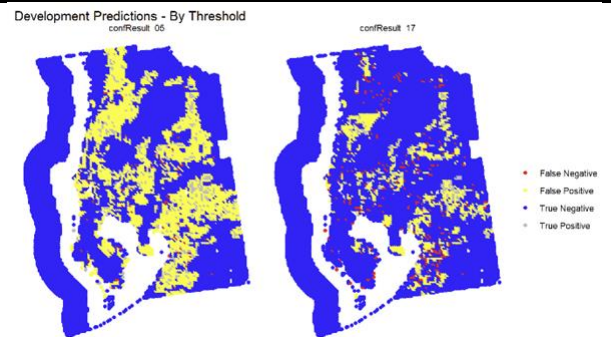
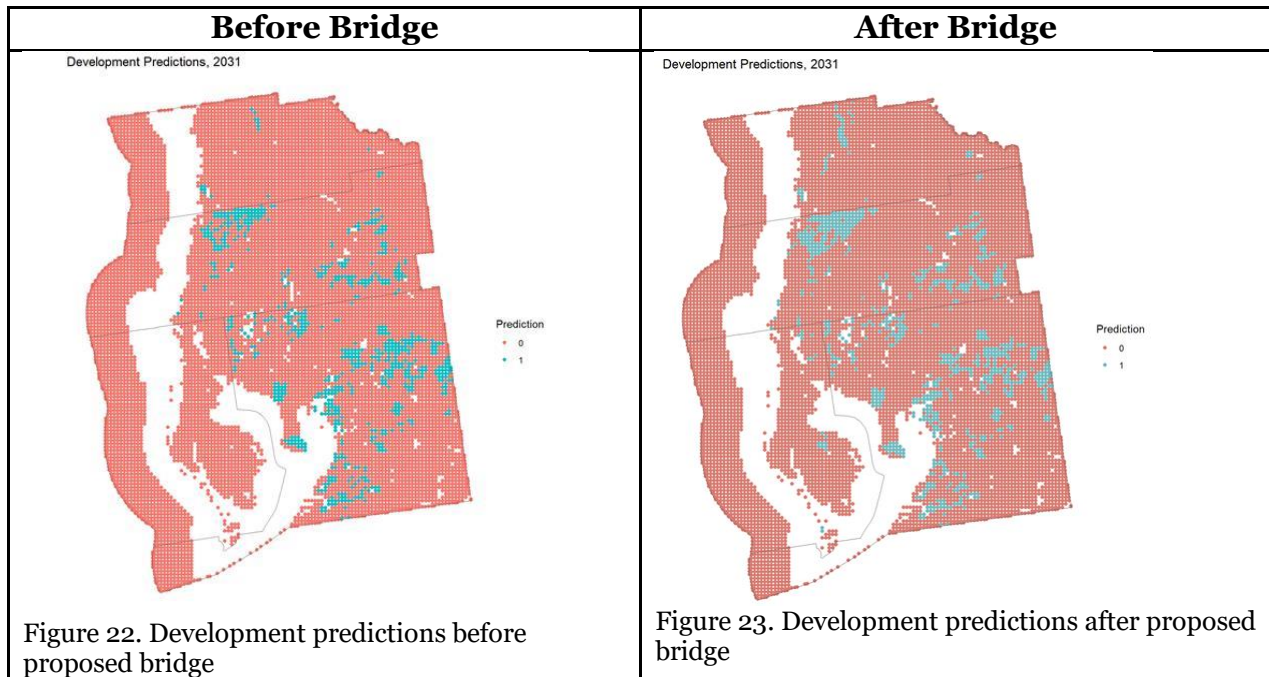


Figure 21. Comparing thresholds mapped

Forecast

`Model6` with a threshold of 0.17 will be used for the forecast, minimizing false positives. To find the impact of the new bridge project, two forecasts were made representing with and without the new bridge. Following the bridge construction, more development is predicted around Ruskin, north of St. Petersburg, and south of Spring Hill.



Assessment of Impact

With the proposed bridge construction, 856 cells representing 0.29mi² is anticipated to convert to developed land. Of that, Hillsborough County is forecasted to have the most land converted at 482 cells or 0.16mi², with Pasco County behind at 309 cells or 0.10mi² converted. Regarding sensitive land, 105 cells of farmland or 0.036mi² and 42 cells of forest or 0.014mi² are expected to convert to developed land.

total_cells	total_area_m	total_km2	total_mi2	
856	770400.4	0.7704004	0.2974531	
Figure 24. Overall land forecasted land conversion				
NAME	total_cells	total_area_m	total_km2	total_mi2
Hernando	44	39600.02	0.0396000	0.0152896
Hillsborough	482	433800.21	0.4338002	0.1674911
Pasco	309	278100.14	0.2781001	0.1073750
Pinellas	21	18900.01	0.0189000	0.0072973
Figure 25. County-by-county forecast				

NAME	Variable	total_cells	total_area_m	total_km2	total_mi2
Hernando	farm	0	0.000	0.0000	0.0000000
Hernando	forest	0	0.000	0.0000	0.0000000
Hernando	otherUndeveloped	4	3600.002	0.0036	0.0013900
Hernando	wetlands	0	0.000	0.0000	0.0000000
Hillsborough	farm	83	74700.037	0.0747	0.0288418
Hillsborough	forest	5	4500.002	0.0045	0.0017375
Hillsborough	otherUndeveloped	7	6300.003	0.0063	0.0024324
Hillsborough	wetlands	0	0.000	0.0000	0.0000000
Pasco	farm	22	19800.010	0.0198	0.0076448
Pasco	forest	37	33300.016	0.0333	0.0128572
Pasco	otherUndeveloped	4	3600.002	0.0036	0.0013900
Pasco	wetlands	0	0.000	0.0000	0.0000000
Pinellas	farm	0	0.000	0.0000	0.0000000
Pinellas	forest	0	0.000	0.0000	0.0000000
Pinellas	otherUndeveloped	0	0.000	0.0000	0.0000000
Pinellas	wetlands	0	0.000	0.0000	0.0000000

Figure 26. Forecast by 2021 land cover type

The changes the model predicts raise concerns about the loss of agricultural land and natural vegetation in the region. These environmental alterations reduce ecosystem services, increase habitat fragmentation, and cause potential long-term impacts on local food production. The increase in development creates more impervious surfaces that elevate runoff levels, affecting plant biodiversity and soil quality. These findings show the need for growth management to find a safe balance between conservation and development.

Key Recommendations

Based on the model predictions and the identified areas of development, three key recommendations are proposed in accordance with the goals of the “Imagine 2040: Tampa Comprehensive Plan” published by the city. First, future growth should be concentrated along major highways, state routes and surface roads to encourage connected development and make use of existing infrastructure. Second, the model and the city goals emphasize the development of high density, mixed-use areas that encompass both existing and expanded transportation features. Third, new development should not be exclusively on environmentally sensitive lands, such as existing farmland and forests, and historically significant areas of Tampa.

Conclusion

The Tampa MSA 2031 development forecast predicts anticipated development and city growth patterns in response to the construction of potential infrastructure. By modeling changes based on previous land use trends and population changes, the model predicts areas where development may change depending on the installation of connecting highway infrastructure. The prediction insights of the model highlight key recommendations that align with the 2040 planning goals of the City of Tampa. This model and its corresponding recommendations support growth that is economically strategic, developmentally focused and environmentally responsible.